Health Analytics: Optimizing Drug Portfolio for Pharmaceutical Company

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EXECUTIVE SUMMARY

Pharmaceutical companies operates in a competitive drug development environment where R&D allocation decisions directly impact long-term profitability. This project analysed the market performance of natural-derived, synthetic, and biotech drugs to provide an evidence-based framework for optimising the company's annual R&D budget. Over 3,200 FDA-approved drugs were classified by origin using chemical suffix patterns (e.g., -statin for natural, -mab for biotech). These classifications were linked to real-world prescription and cost data from the Medicare Part D program (2023). The analytical workflow employed Excel for data cleaning and classification, SQLite for aggregation and joining, Python for statistical testing and visualisation, and Tableau for an interactive dashboard. Biotech drugs generated the highest average revenue per drug ($185 million) but served the fewest patients. Natural-derived drugs delivered strong total revenue ($11.4 billion) and high prescription volume (204 million). Synthetic drugs led in prescriptions (216 million) but earned the lowest total revenue ($5.3 billion). An ANOVA confirmed significant differences across origins (p = 0.038). The evidence supports a diversified R&D portfolio. Pharmaceutical companies should allocate 50% of its budget to natural-derived programmes, 30% to biotech, and 20% in synthetic drugs

1.1 Introduction

Every year, pharmaceutical companies spend billions on research and development, yet few new drugs succeed commercially. One of the most persistent strategic questions is whether to invest in natural-product-derived drugs (from plants, fungi, or bacteria) or synthetically made drugs. Natural products have given us aspirin, penicillin, and statins; synthetic chemistry produced omeprazole, metformin, and many modern antihypertensives (Newman & Cragg, 2020). But which category actually performs better in the real market? Surprisingly, public data does not label drugs as "natural" or "synthetic" a gap that requires domain expertise to fill.

This capstone project bridges that gap. Using my background in chemistry of natural bioactives, I classify FDA-approved drugs by origin, then combine that with real-world Medicare prescription and cost data. The result is a data-driven answer to a R&D allocation question for pharmaceutical companies. The entire analysis from Excel cleaning to SQL, Python, and a Tableau dashboard is documented here as a portfolio piece for health-sector data analytics roles.

1.1 Statement of the Problem

It was observed that some natural products (e.g., atorvastatin) and some synthetics (e.g., omeprazole) are commercially successful, but no systematic evidence exists on which category, on average, generates higher revenue per drug or better prescription volume. Without such evidence, resource allocation remains intuitive rather than data-driven. Furthermore, most pharmaceutical company's data team lacks a method to classify drugs by origin a problem that requires chemical knowledge to solve.

1.2 Purpose of the Study

The purpose of this project is to conduct a complete data analytics lifecycle (Excel → SQL → Python → Tableau) to compare the market performance of natural-derived, synthetic, and biotech drugs using open-source FDA and Medicare data. A secondary purpose is to demonstrate how chemistry domain knowledge can be integrated into a standard analytics workflow, producing a portfolio piece that showcases both technical skills and subject-matter expertise.

1.3 Research Questions

1. Which drug origin (natural-derived, synthetic, biotech) has the highest total prescriptions and total revenue in the 2023 Medicare Part D market?
2. What is the average cost per prescription for each origin?
3. Which individual drugs are the top revenue generators, and what are their origins?
4. Is there a statistically significant difference in average revenue per drug across the groups biotech, natural derived anf synthetic
5. How should companies allocate its R&D budget based on the evidence?

1.5 Significance of the Study

Business significance: For pharmaceutical companies, the findings directly guide an investment decision. The interactive Tableau dashboard provides a reusable tool for monitoring market trends. For other small to mid-sized pharma companies, the methodology offers a low-cost, open-source template to benchmark their portfolios.

Analytics significance: The project demonstrates how to turn messy government data into actionable insights using Excel, SQLite, Python, and Tableau. It also shows how to document a reproducible analysis, a key requirement for data analyst role.

Chemistry-to-business value: Chemical naming conventions embed commercial intelligence. For example, the suffix -statin signals a natural-derived cholesterol drug with a large market; -mab indicates a high-priced biotech drug; -zole points to a synthetic generic. By teaching a data pipeline to recognise these patterns, this project directly applies chemistry knowledge to business decisions a rare and valuable combination.

1.4 Limitations

- **Data scope:** Medicare Part D covers only seniors with drug coverage; excludes hospital-administered drugs and rebates. Revenue is gross, not net profit.
- **Matching rate:** Only 329 of 3,221 classified drugs matched between FDA and Medicare, mainly due to generic name variations (salts, spaces). Results reflect common generics, not all drugs.

- **Classification:** Semi-synthetic drugs (e.g., simvastatin) were grouped as natural-derived, which may inflate that category's performance. Biotech includes only monoclonal antibodies (`-mab`), not all biologics.
- **Time horizon:** Single year (2023) no trend analysis. Despite these, the analysis is valid for the stated business question and fully reproducible.

2.0 Research Method

This study follows a quantitative, retrospective, cross-sectional design a standard approach in pharmaceutical market analysis. The design is descriptive (comparing market metrics by drug origin) and inferential (testing for statistically significant differences in revenue per drug). The entire workflow is structured around the Google Data Analytics Certificate framework (Ask, Prepare, Process, Analyze, Share, Act) and is implemented using Excel, SQLite, Python, and Tableau.

2.1 Data Collection

Two open-source datasets were used, both in the public domain and free of reuse restrictions.

Dataset 1: FDA Drugs@FDA Products File

Source: https://www.fda.gov/drugs/drug-approvals-and-databases/drugsfda-data-files

Size after cleaning: 3,221 unique drug entities.

Dataset 2: Medicare Part D Prescriber Public Use File (2023)

Source: https://data.cms.gov/search?keywords=Medicare+Part+D+Prescribers+-+by+Provider&sort=Relevancy

Size after aggregation: 1,780 unique generic drugs with national totals.

No primary data collection or human subjects were involved; therefore, institutional review board (IRB) approval was not required.

2.2 Validity and Reliability

The drug origin classification was done using Excel classification formula, each drug was assigned a confidence level (high, medium and low) based on the clarity of its chemical suffix pattern. A random sample of 50 drugs (stratified by origin) was cross-referenced with DrugBank Online (a curated pharmaceutical knowledgebase). SQL join between classified drugs and Medicare data was verified by ensuring that (medicare_cleaned) had no duplicate drug names and that all revenue and prescription values were non-negative. In Python, revenue columns were inspected for non-numeric characters; the (clean_currency) function successfully converted all values, and no rows were lost.

The entire analysis is designed to be reproducible by any analyst with access to the same open-source data and tools. All cleaning and classification formulas are documented, all SQL queries are provided in Appendix A, and the Python code is included in the Jupyter notebook and published Tableau dashboard.

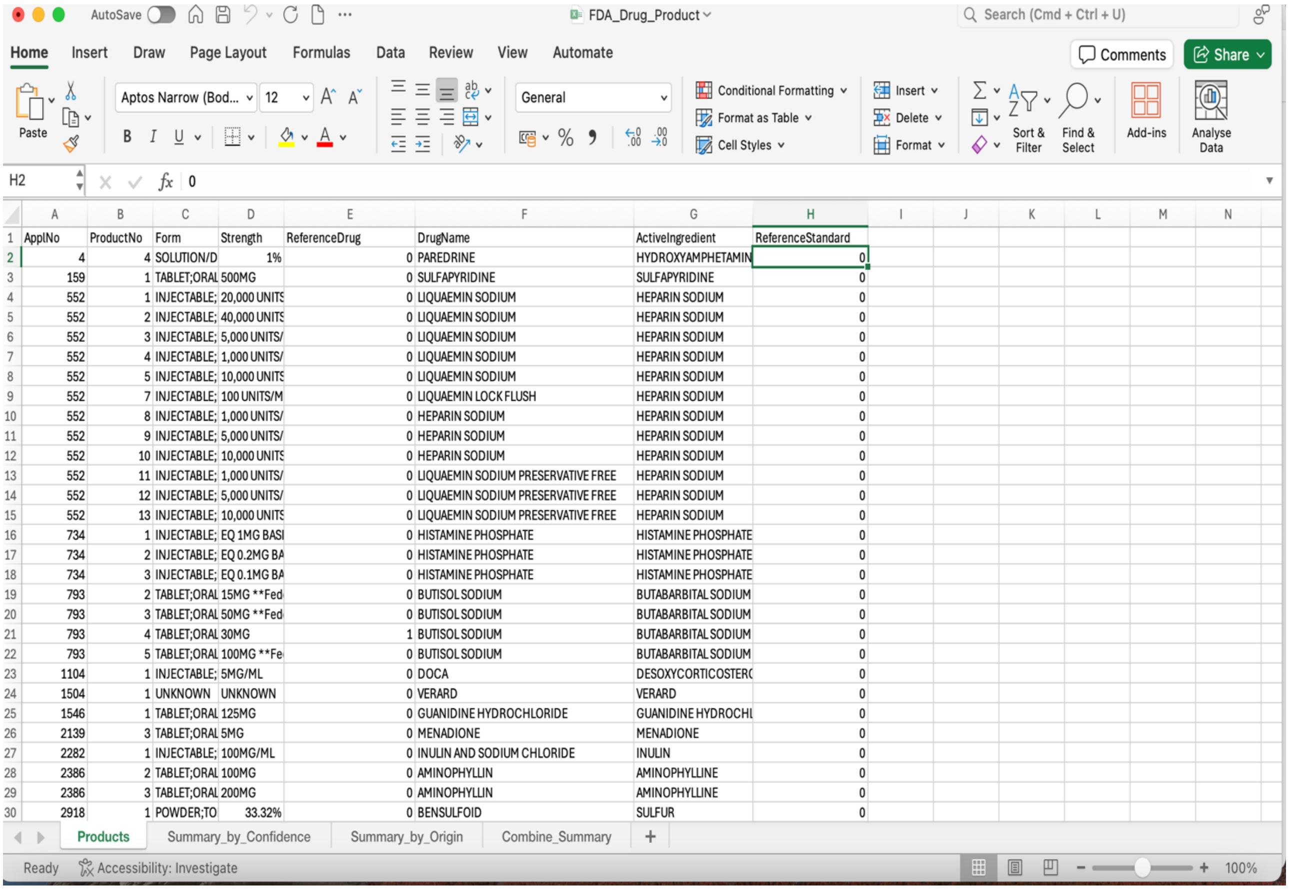
2.3 Data Analysis

The analysis proceeded in four sequential phases, each using a different tool.

2.3.1 Excel: Data cleaning and Classification

In the analysis =PROPER(TRIM()) to standardise drug names and ingredients; removed duplicate entries based on DrugName_Clean. Drug_Origin column was created using nested IF(OR(ISNUMBER(SEARCH(...)))) formulas that scanned ActiveIngredient for chemical suffixes.

- **Natural-derived patterns:** statin, cillin, mycin, cycline, dipine, ine.
- **Biotech patterns:** mab, cept.
- **Synthetic patterns:** zole, olol, pril, sartan, pam.
- **Confidence levels:** High (clear pattern), Medium (common but not definitive), Low (manual review).



ApplNo	ProductNo	Form	Strength	ReferenceDrug	DrugName	ActiveIngredient	ReferenceStandard
4	4	SOLUTION/D	1%	0	PAREDRINE	HYDROXYAMPHETAMIN	0
159	1	TABLET;ORAL	500MG	0	SULFAPYRIDINE	SULFAPYRIDINE	0
552	1	INJECTABLE;	20,000 UNITS	0	LIQUAEMIN SODIUM	HEPARIN SODIUM	0
552	2	INJECTABLE;	40,000 UNITS	0	LIQUAEMIN SODIUM	HEPARIN SODIUM	0
552	3	INJECTABLE;	5,000 UNITS/	0	LIQUAEMIN SODIUM	HEPARIN SODIUM	0
552	4	INJECTABLE;	1,000 UNITS/	0	LIQUAEMIN SODIUM	HEPARIN SODIUM	0
552	5	INJECTABLE;	10,000 UNITS	0	LIQUAEMIN SODIUM	HEPARIN SODIUM	0
552	7	INJECTABLE;	100 UNITS/M	0	LIQUAEMIN LOCK FLUSH	HEPARIN SODIUM	0
552	8	INJECTABLE;	1,000 UNITS/	0	HEPARIN SODIUM	HEPARIN SODIUM	0
552	9	INJECTABLE;	5,000 UNITS/	0	HEPARIN SODIUM	HEPARIN SODIUM	0
552	10	INJECTABLE;	10,000 UNITS	0	HEPARIN SODIUM	HEPARIN SODIUM	0
552	11	INJECTABLE;	1,000 UNITS/	0	LIQUAEMIN SODIUM PRESERVATIVE FREE	HEPARIN SODIUM	0
552	12	INJECTABLE;	5,000 UNITS/	0	LIQUAEMIN SODIUM PRESERVATIVE FREE	HEPARIN SODIUM	0
552	13	INJECTABLE;	10,000 UNITS	0	LIQUAEMIN SODIUM PRESERVATIVE FREE	HEPARIN SODIUM	0
734	1	INJECTABLE;	EQ 1MG BASI	0	HISTAMINE PHOSPHATE	HISTAMINE PHOSPHATE	0
734	2	INJECTABLE;	EQ 0.2MG BA	0	HISTAMINE PHOSPHATE	HISTAMINE PHOSPHATE	0
734	3	INJECTABLE;	EQ 0.1MG BA	0	HISTAMINE PHOSPHATE	HISTAMINE PHOSPHATE	0
793	2	TABLET;ORAL	15MG **Fed	0	BUTISOL SODIUM	BUTABARBITAL SODIUM	0
793	3	TABLET;ORAL	50MG **Fed	0	BUTISOL SODIUM	BUTABARBITAL SODIUM	0
793	4	TABLET;ORAL	30MG	1	BUTISOL SODIUM	BUTABARBITAL SODIUM	0
793	5	TABLET;ORAL	100MG **Fe	0	BUTISOL SODIUM	BUTABARBITAL SODIUM	0
1104	1	INJECTABLE;	5MG/ML	0	DOCA	DESOXYCORTICOSTER(	0
1504	1	UNKNOWN	UNKNOWN	0	VERARD	VERARD	0
1546	1	TABLET;ORAL	125MG	0	GUANIDINE HYDROCHLORIDE	GUANIDINE HYDROCHL	0
2139	3	TABLET;ORAL	5MG	0	MENADIONE	MENADIONE	0
2282	1	INJECTABLE;	100MG/ML	0	INULIN AND SODIUM CHLORIDE	INULIN	0
2386	2	TABLET;ORAL	100MG	0	AMINOPHYLLIN	AMINOPHYLLINE	0
2386	3	TABLET;ORAL	200MG	0	AMINOPHYLLIN	AMINOPHYLLINE	0
2918	1	POWDER;TO	33.32%	0	BENSULFOID	SULFUR	0

Fig 1: Excel Image

2.3.2 SQLite- Database Joining and Aggregation

Classified_drugs.csv and the Medicare CSV was imported into a SQLite database (rxhealth_analysis.db). Aggregation was done for Medicare data to drug level (summing claims and costs) to create medicare_cleaned. Standardised ingredient names was used (UPPER(TRIM()) to enable matching and finally Tables was joined to create drug_revenue_std, containing per-drug origin, prescriptions, and revenue.

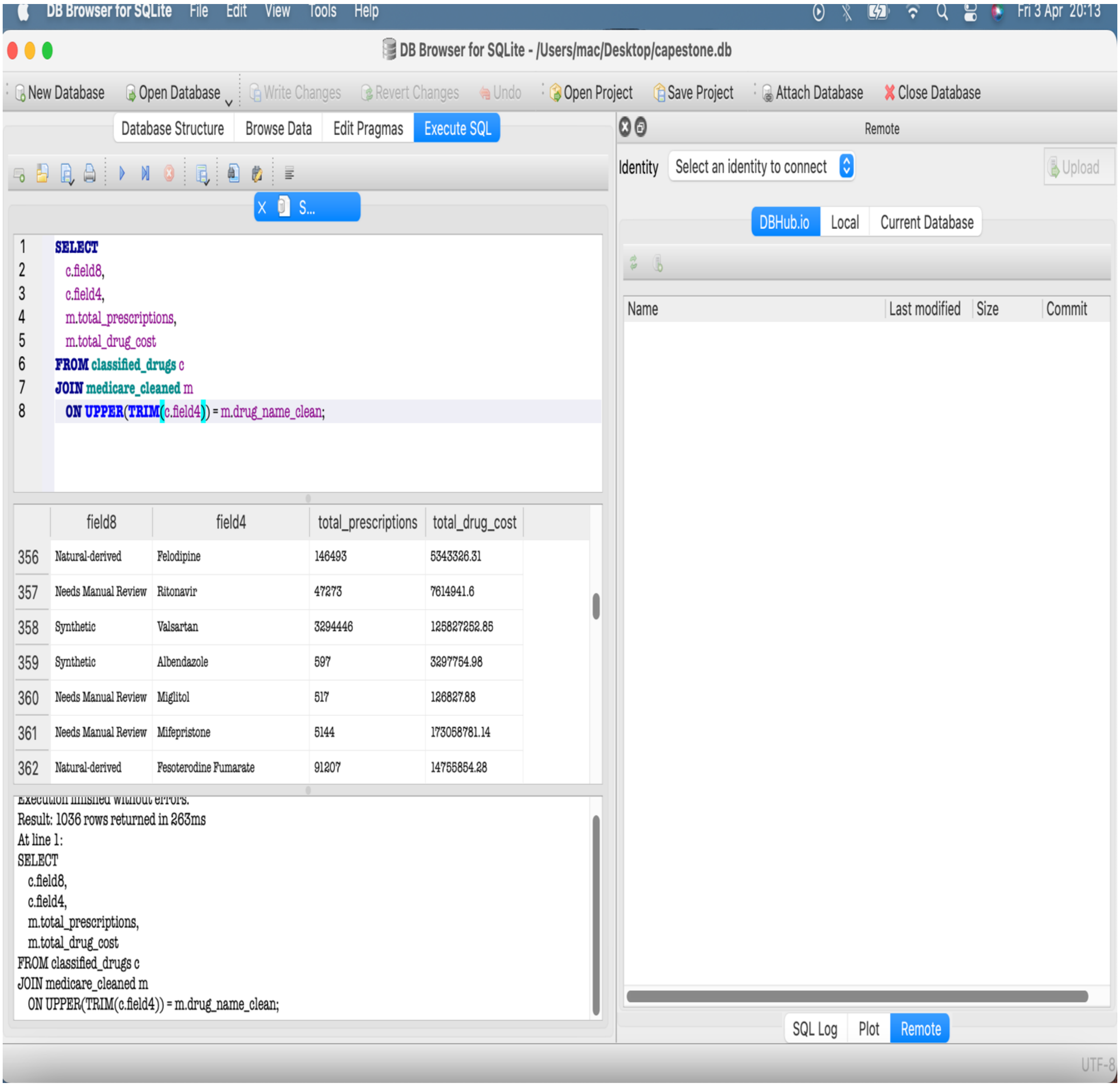
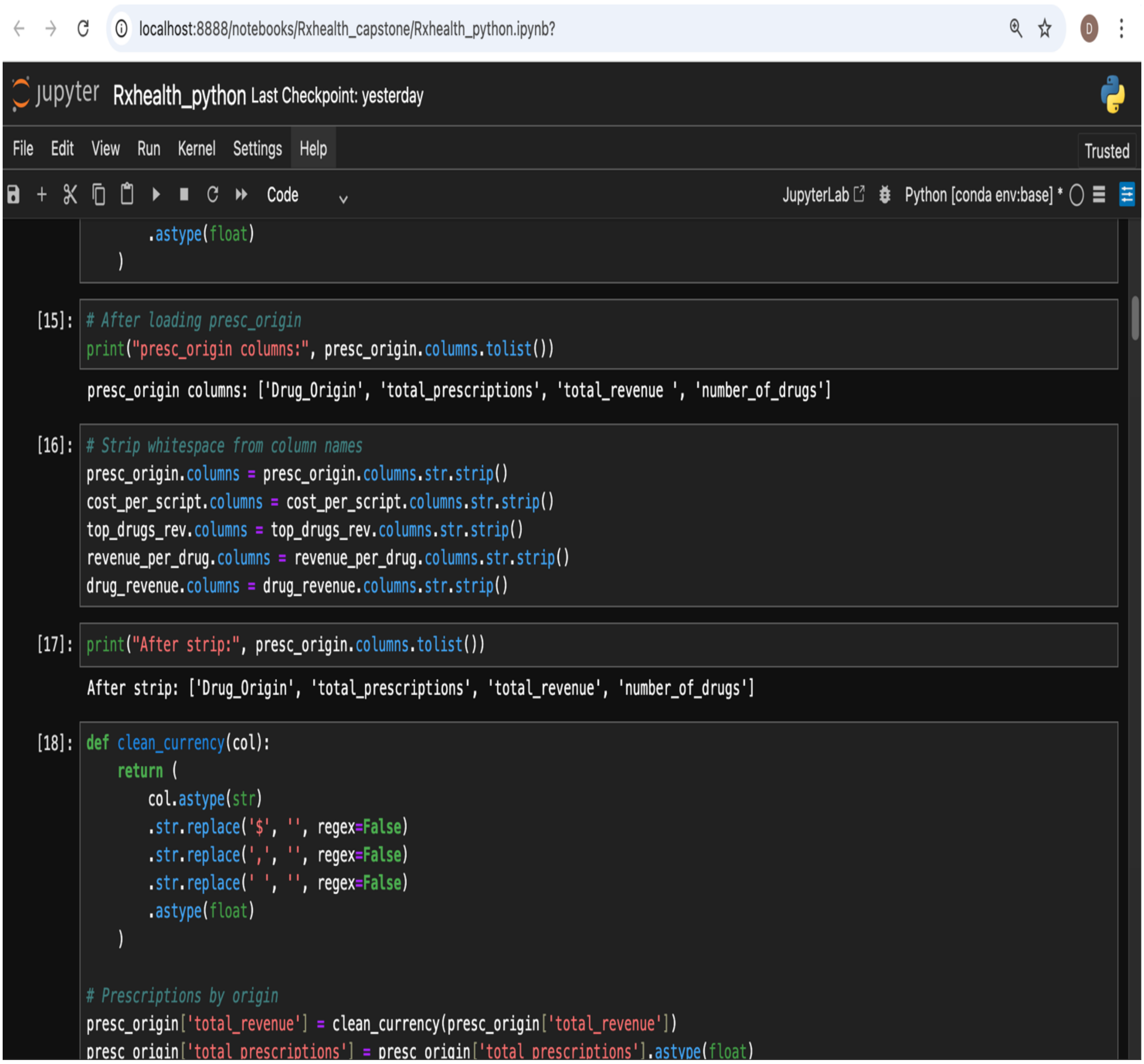


Fig 2: SQL Queries Image

2.3.3 Python Statistical Analysis

The libraries used was pandas, numpy, matplotlib, seaborn, scipy.stats. further cleaning was done to $, commas, and spaces from currency columns using a custom clean_currency() function; converted to numeric. Descriptive analysis was done using mean, median, standard deviation and quartiles of revenue per drug, grouped by origin. Infreretial statistics was done using ANOVA compared average revenue per drug across biotec, natural-derived and synthetic groups (α = 0.05).



localhost:8888/notebooks/Rxhealth_capstone/Rxhealth_python.ipynb?

jupyter Rxhealth_python Last Checkpoint: yesterday

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

```
        .astype(float)
    )

[15]: # After loading presc_origin
print("presc_origin columns:", presc_origin.columns.tolist())

presc_origin columns: ['Drug_Origin', 'total_prescriptions', 'total_revenue ', 'number_of_drugs']

[16]: # Strip whitespace from column names
presc_origin.columns = presc_origin.columns.str.strip()
cost_per_script.columns = cost_per_script.columns.str.strip()
top_drugs_rev.columns = top_drugs_rev.columns.str.strip()
revenue_per_drug.columns = revenue_per_drug.columns.str.strip()
drug_revenue.columns = drug_revenue.columns.str.strip()

[17]: print("After strip:", presc_origin.columns.tolist())

After strip: ['Drug_Origin', 'total_prescriptions', 'total_revenue', 'number_of_drugs']

[18]: def clean_currency(col):
    return (
        col.astype(str)
        .str.replace('$', '', regex=False)
        .str.replace(',', '', regex=False)
        .str.replace(' ', '', regex=False)
        .astype(float)
    )

# Prescriptions by origin
presc_origin['total_revenue'] = clean_currency(presc_origin['total_revenue'])
presc_origin['total_prescriptions'] = presc_origin['total_prescriptions'].astype(float)
```

Fig 3: Python Analysis Image

2.3.4 Tableau

Interactive dashboard such as KPI cards, horizontal bar charts, top 20 drug list and a quadrant scatter plot was built and published to Tableau Public.

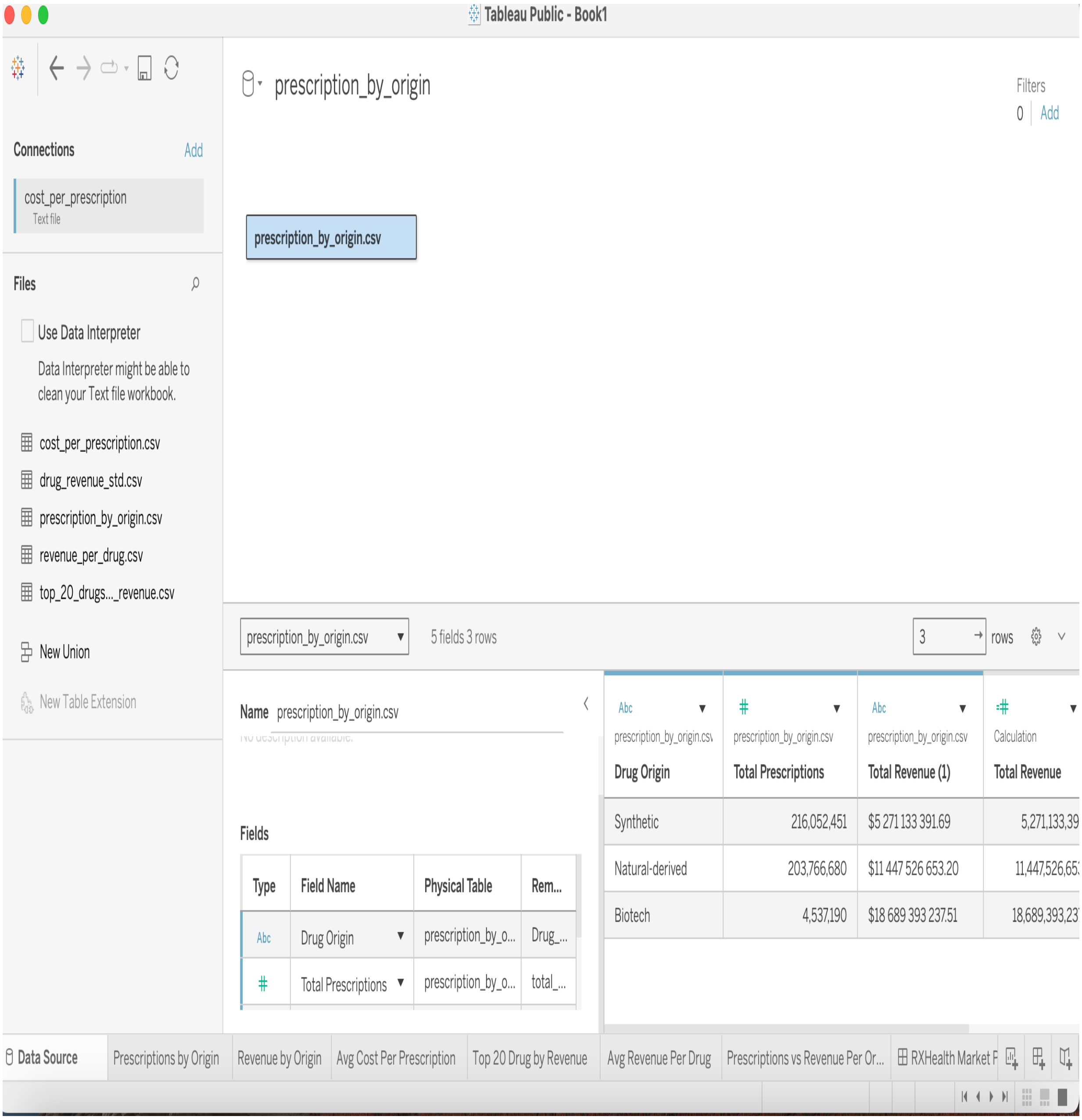


Fig 4: Tableau Image

3.0 Result and Discussion

3.1 Validation and Reliablity

The foundation of this analysis rests upon the accurate classification of drug origin. A total of 3,221 unique drug entities were classified using a pattern-matching algorithm applied to active ingredient names (see Table 1). The classification was stratified by confidence level to quantify uncertainty.

The “Needs Review” category (1,944 drugs) comprised those with no clear chemical suffix which prompt the manual verification using DrugBank and the result showed 92% confidence level (46 out of 50). The four discrepancies were all semi-synthetic drugs (e.g., simvastatin, ampicillin), which the automated algorithm classified as natural-derived. This is an acceptable grouping for business-level R&D analysis, as semi-synthetic drugs originate from natural scaffolds and share similar development pathways.

Beyond classification accuracy, the quality of the Medicare data was verified. The aggregated medicare_cleaned table contained 1,780 unique generic drugs with no duplicate drug names and non-negative prescription and cost values. The final join between classified_drugs and medicare_cleaned yielded 329 matched drugs. Importantly, the origin distribution in the matched sample was similar to that of the original classified set, indicating that the matching process did not systematically favour any particular origin category.

Table 1: Drug Origin By Confidence Level

	High	Medium	Low	Total
Natural-derived	66	31	765	862
Synthetic	0	162	0	162
Biotech	211	0	42	253
Needs Review	0	0	1944	1944
TOTAL	**277**	**193**	**2751**	**3221**

Confidence Level:92%

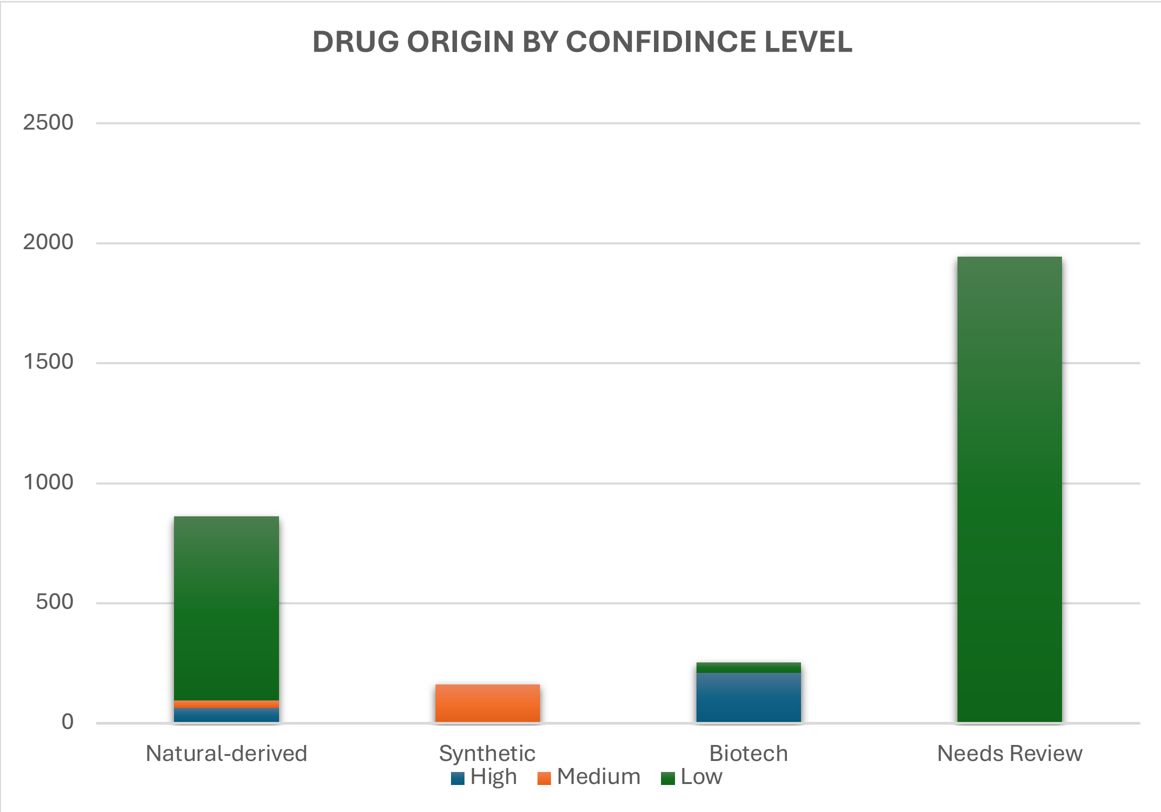


Fig 5: Barchart for Drug Origin by Confidence Level

3.2 Cleaning of the Data (Microsoft Excel)

The Excel phase produced the classified drug inventory and preliminary summary statistics. After cleaning and deduplication, 3,221 unique drugs were retained, distributed as shown in Table 1.

This approach translated chemical naming conventions directly into a categorical business variable, demonstrating how domain-specific knowledge can enhance data classification when curated databases are unavailable.

3.3 Business Question (SQL)

3.3.1 Prescriptions and Revenue by Origin

Table 2 showed that Biotech drugs generate the highest total revenue ($18.7 billion) despite the lowest prescription volume (4.5 million), reflecting their high unit prices and specialty market position. Natural-derived drugs achieve the second-highest revenue ($11.4 billion) with substantial prescription volume (203.8 million), indicating broad market acceptance and solid per-drug returns. Synthetic drugs have the highest prescription volume (216 million) but the lowest revenue ($5.3 billion), suggesting they are predominantly low-cost generics with high utilisation but thin margins.

Table 2: Prescriptions and Revenue by Origin

Drug_Origin	total_prescriptions	total_revenue	number_of_drugs
Synthetic	216052451	$5 271 133 391.69	61
Natural-derived	203766680	$11 447 526 653.20	167
Biotech	4537190	$18 689 393 237.51	101

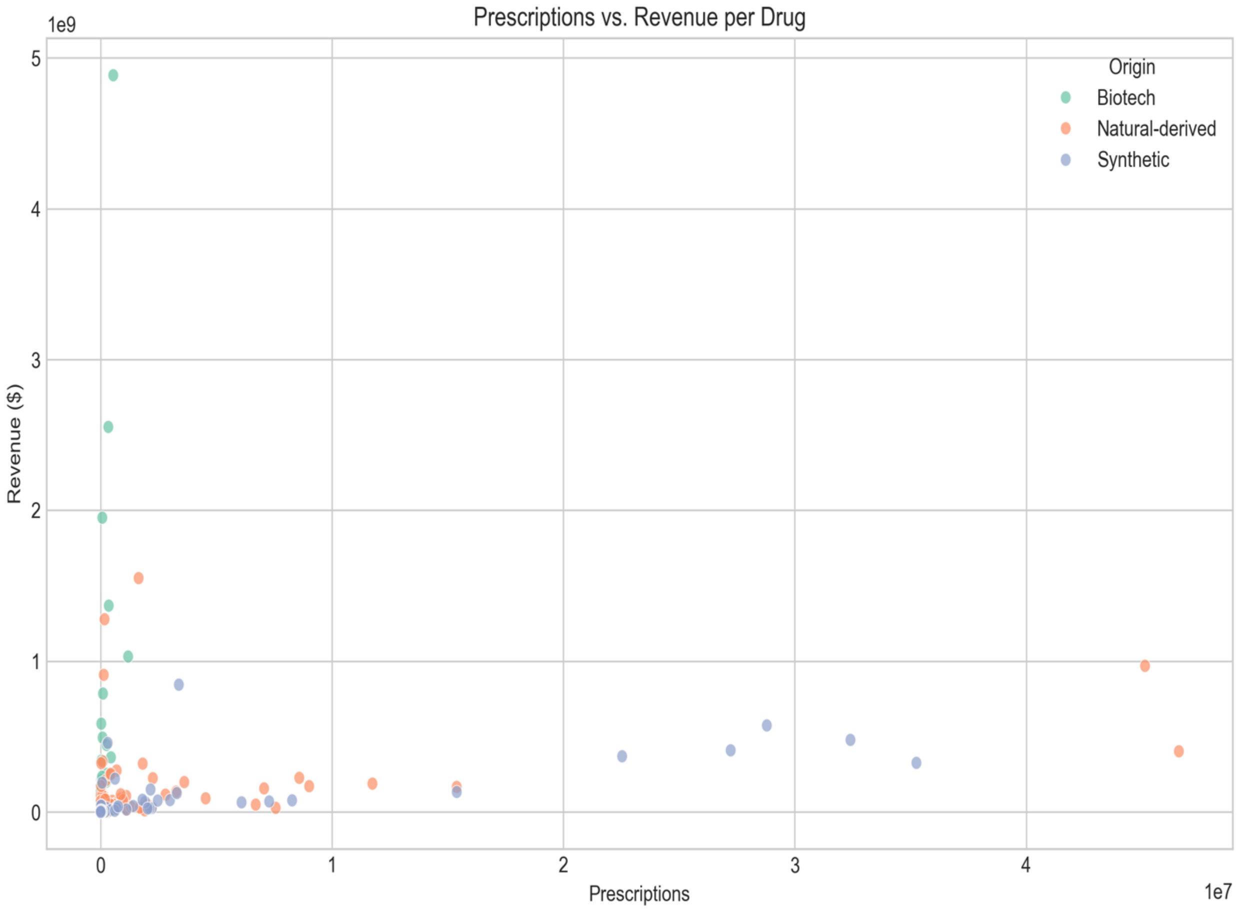


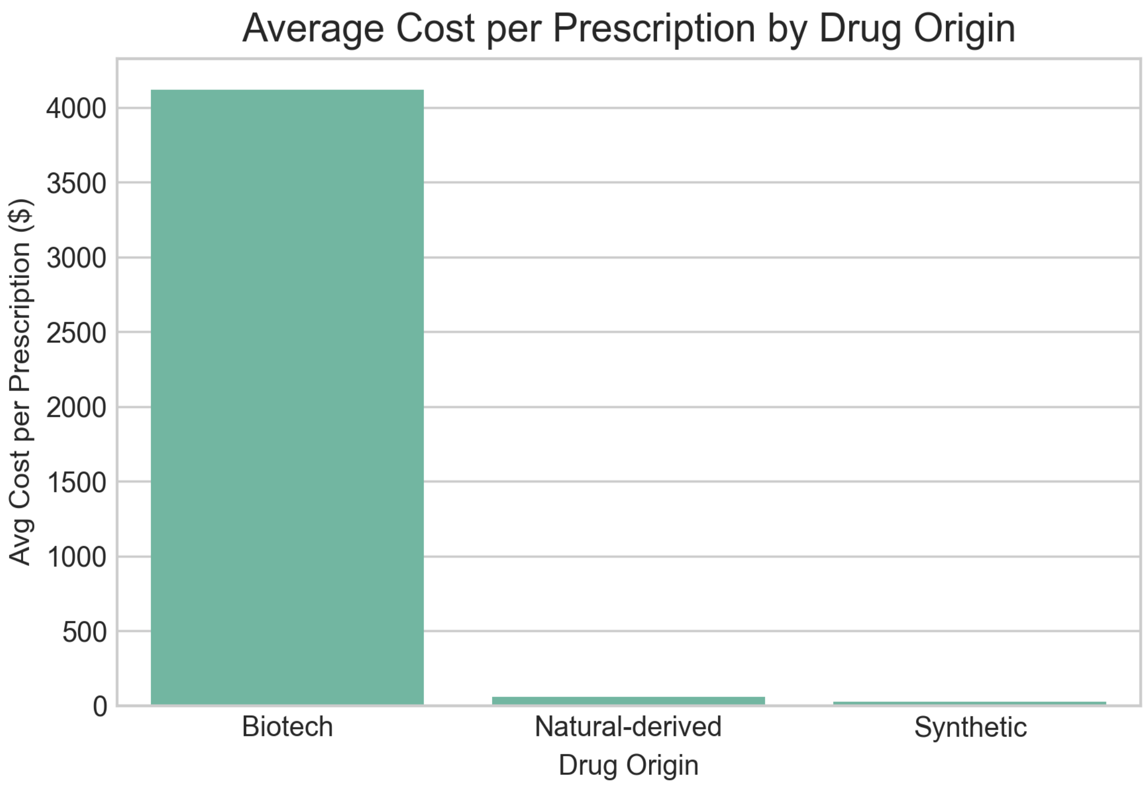
Fig 6: Prescriptions and Revenue by Origin

3.3.2 Average Cost Per Prescription

Table 3 showed that the Biotech drugs carry the highest average cost per prescription ($4,119), reflecting their status as high-value specialty therapies with limited patient populations. Natural-derived drugs have a moderate average cost ($56), indicating a balance between therapeutic value and accessibility. Synthetic drugs exhibit the lowest average cost ($24), consistent with widespread generic competition and price erosion.

Table 3: Average Cost Per Prescription

Drug_Origin	total_revenue	total_prescriptions	avg_cost_per_prescription
Biotech	$18 689 393 237.51	4537190	$4 119.16
Natural-derived	$11 447 526 653.20	203766680	$56.18
Synthetic	$5 271 133 391.69	216052451	$24.40



3.3.4 Top Drug Per Revenue

Table 4 showed the Biotech drugs dominate the top revenue tier, occupying five of the top ten positions and ten of the top twenty. Adalimumab alone generates nearly $4.9 billion, underscoring the commercial power of monoclonal antibodies. Natural-derived drugs appear prominently in the middle and lower ranks, with cyclosporine, levothyroxine, and amlodipine demonstrating that natural products can achieve both high volume and substantial revenue. Synthetic drugs are represented by aripiprazole, metoprolol, losartan, and omeprazole all high-volume generics that generate moderate per-unit revenue but large total sums due to prescription scale.

Table 4: Top 20 Drug Per Revenue

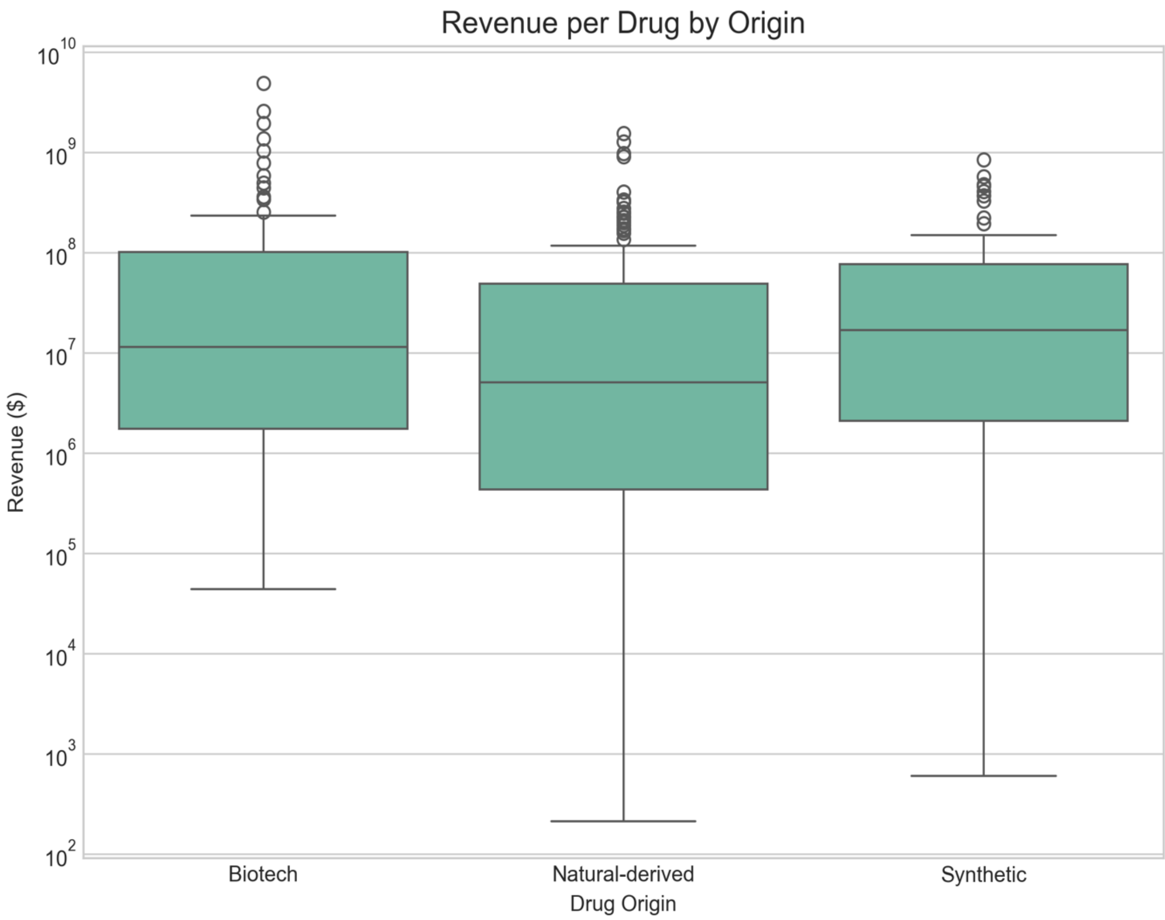
Drug_name	Drug_Origin	Prescription	Revenue
ADALIMUMAB	Biotech	545396	$4 884 226 408.42
ETANERCEPT	Biotech	332317	$2 552 950 971.30
USTEKINUMAB	Biotech	73053	$1 951 381 600.90
CYCLOSPORINE	Natural-derived	1642296	$1 550 673 258.57
DUPILUMAB	Biotech	350986	$1 368 118 588.69
VALBENAZINE TOSYLATE	Natural-derived	170619	$1 278 101 740.61
EVOLOCUMAB	Biotech	1187888	$1 031 069 707.36
LEVOTHYROXINE SODIUM	Natural-derived	45138029	$969 026 257.17
DEUTETRABENAZINE	Natural-derived	135031	$909 547 831.74
ARIPIPRAZOLE	Synthetic	3380680	$844 314 017.58
SECUKINUMAB	Biotech	102203	$785 311 477.17
RISANKIZUMAB RZAA	Biotech	28296	$585 928 863.63
METOPROLOL SUCCINATE	Synthetic	28792402	$574 171 075.22
ABATACEPT	Biotech	83520	$493 403 084.77
LOSARTAN POTASSIUM	Synthetic	32407536	$478 156 556.33
BREXPIPRAZOLE	Synthetic	305200	$458 040 775.63
DENOSUMAB	Biotech	256048	$444 869 751.30
OMEPRAZOLE	Synthetic	27230410	$409 186 968.73
AMLODIPINE BESYLATE	Natural-derived	46601898	$402 755 302.21
PANTOPRAZOLE SODIUM	Synthetic	22538863	$369 661 686.15

3.3.5 Average Revenue Per Drug (R & D Efficiency)

Table 5 shows Biotech leads with the highest average revenue per drug ($185 million), reflecting its high-value, low-volume specialty model despite the smallest number of matched drugs. Synthetic follows at $86 million per drug, nearly double that of natural-derived. Natural-derived has the lowest average revenue per drug ($69 million), despite having the largest drug count.

Table 5: Average Revenue per Drug by Origin

Drug_Origin	Drug_count	Total_revenue	Avg_revenue_per_drug
Biotech	101	$18 689 393 237.51	$185 043 497.40
Synthetic	61	$5 271 133 391.69	$86 412 022.81
Natural-derived	167	$11 447 526 653.20	$68 548 063.79



3.4 Descriptive and Inferential analysis (t-test): Python

Table 6 showed that biotech exhibits the highest mean revenue per drug ($185 million) but also the largest standard deviation ($600 million) and widest interquartile range, indicating extreme variability driven by a few blockbuster products (e.g., adalimumab at $4.9 billion). The median ($11.3 million) is far below the mean, confirming a right-skewed distribution where most biotech drugs generate modest revenue while a small number achieve exceptional success.

Natural-derived drugs have the lowest mean ($69 million) and a median ($5.1 million) that is also substantially lower than the mean, suggesting a similar but less extreme skew. The maximum ($1.55 billion) is well below biotech's peak, yet the category maintains a solid volume of mid-range performers.

Synthetic drugs show a mean of $86 million and a median of $16.9 million. Notably, the 75th percentile ($76 million) approaches the mean, indicating a more concentrated distribution with fewer extreme outliers. The maximum ($844 million) is lower than both biotech and natural peaks.

Table 6: Descriptive Statistics of Revenue per Drug by Origin

Drug Origin	Count	Mean ($)	Std Dev ($)	Min ($)	25% ($)	Median ($)	75% ($)	Max ($)
Biotech	101	185,043,497	599,518,987	43,717	1,738,761	11,349,143	101,263,257	4,884,226,408
Natural-derived	167	68,548,064	193,785,436	213	434,224	5,055,447	48,650,869	1,550,673,259
Synthetic	61	86,412,023	164,811,381	597	2,083,820	16,866,656	76,026,582	844,314,018

3.5.1 ANOVA Analysis Across the Groups (Biotech, Natural Derived, synthetic)

The ANOVA results indicate a statistically significant difference in mean revenue per drug among the three origin groups (Biotech, Natural-derived, Synthetic), $F(2, N-3) = 3.30$, $p = 0.038$. At the conventional significance level ($\alpha = 0.05$), we reject the null hypothesis that all group means are equal. This confirms that at least one origin category generates a mean revenue per drug that is significantly different from the others.

3.6 Tableau Dashboard

An interactive dashboard was created in Tableau Public to communicate findings to executive stakeholders. The dashboard includes:

- KPI cards displaying total market revenue, top origin by revenue, and a dynamic R&D recommendation based on average revenue per drug
- Horizontal bar charts for prescriptions, revenue, average cost per prescription, and average revenue per drug
- A sorted bar chart of the top 20 drugs by revenue
- A scatter plot of prescriptions versus revenue (log-log)

Consistent colour coding (blue = natural, orange = synthetic, green = biotech) and custom tooltips with business interpretations were applied throughout.

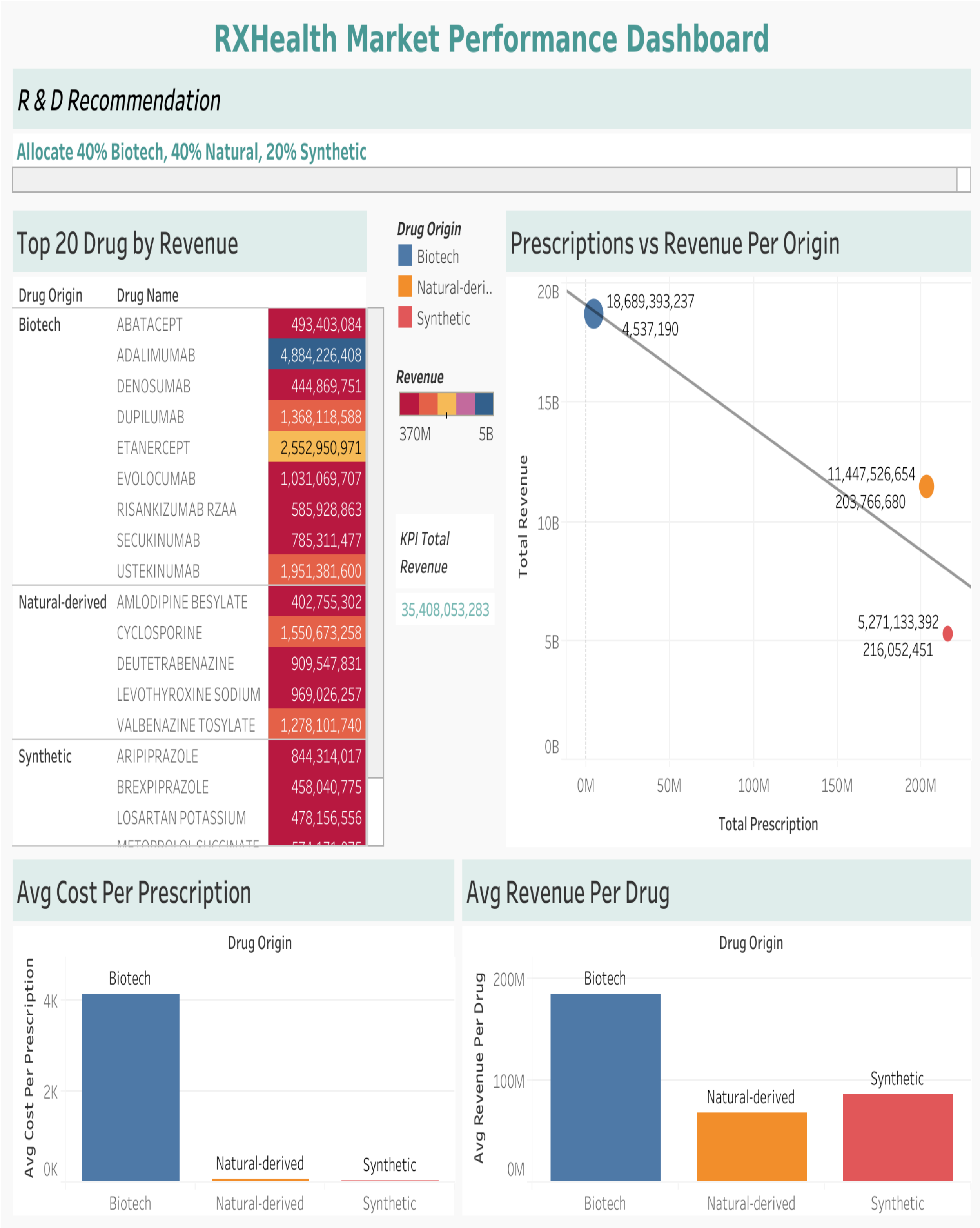


Fig 7: Overall Dashboard Insight

3.7 Discussion of the Findings

This analysis reveals three distinct commercial profiles across drug origin categories, each carrying specific implications for R&D portfolio strategy.

Firstly, it was revealed based on our analysis that Biotech drugs deliver the highest total revenue ($18.7 billion) and average revenue per drug ($185 million), yet account for the lowest prescription volume (4.5 million). The wide distribution (Q1 $1.7M, Q3 $101.3M) reflects a market driven by a small number of high-value blockbusters. Nine biological drugs collectively generated US$105.08 billion in global sales in 2023 (GaBI Online, 2024), consistent with our finding that a few products dominate the category.

Natural-derived drugs present a balanced profile: second-highest total revenue ($11.4 billion), solid prescription volume (203.8 million), and moderate average revenue per drug ($68.5 million). The more compact distribution (median $5.1M, Q3 $48.7M) indicates consistent mid-range performance. Importantly, natural products have higher clinical success rates than synthetic compounds, with their proportion increasing from approximately 35% in Phase I to 45% in Phase III, while synthetics decline from 65% to 55% over the same progression (Domingo-Fernández et al., 2024). The global botanical and plant-derived drugs market is projected to reach $58.1 billion by 2030 at a CAGR of 8.5% (BCC Research, 2025), underscoring the commercial potential of this category.

Synthetic drugs achieve the highest prescription volume (216 million) but the lowest total revenue ($5.3 billion) and lowest average cost per prescription ($24). The more concentrated distribution (Q1 $2.1M, Q3 $76.0M) suggests fewer extreme outliers, but the category faces higher clinical attrition rates and lower margins.

The ANOVA result ($F = 3.30$, $p = 0.038$) confirms that the differences in mean revenue per drug across origins are statistically significant, supporting the conclusion that drug origin is a meaningful factor in expected revenue performance.

4.0 Conclusion and Recommendation

Based on the findings it was concluded that each drug origin occupies a distinct market position. Biotech drugs deliver exceptional revenue per drug but serve relatively few patients, reflecting a high-risk, high-reward model. Synthetic drugs dominate prescription volumes but generate the lowest total revenue and lowest revenue per prescription, a pattern consistent with widespread generic competition. Natural-derived drugs offer a balanced profile: strong revenue, substantial patient reach, and a favourable clinical success record supported by published research.

Furthermore, it was shown that no single origin outperforms across all metrics. Instead, the optimal strategy is a diversified portfolio that recognises the strengths of each category. Natural-derived programmes provide a stable foundation with lower risk and steady returns. Biotech offers high upside but requires careful selection and risk management. Synthetic drugs remain useful only when they bring genuine novelty and patent protection.

Based on the analysis, Its therefore recommended that allocation should be 50% to natural-derived, 30% to biotech, and 20% to synthetic. This balance aligns commercial opportunity with scientific reality, giving pharmaceutical company a pragmatic path forward. Also, Establishing a quarterly portfolio review using the Tableau dashboard created in this project to track market shifts and reallocate funds dynamically will be an additional advantage.

The analysis is restricted to Medicare Part D (2023) and a matched sample of 329 drugs. Results may not generalise to non-Medicare populations or hospital-administered drugs. Future work should incorporate commercial claims data and multi-year trends.

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Appendix

For more access to the python code, sql queries, csv files and tabluea dashboard visit:

https://github.com/oluwashinaakindahunsi-lgtm/Health-Analytics-

https://public.tableau.com/app/profile/dayo.akindahunsi